Reassessing the Walsh insulin dosing constants in type 1 diabetes

Age composition in entered settings, and the limits of measurement from device data

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Abstract

Insulin pump therapy is initialised from three rules that divide a constant by total daily dose: 1700 for the sensitivity factor, 500 for the carbohydrate ratio, and half the daily dose for basal. The rules predate continuous glucose monitoring and have been validated mostly in small cohorts of mixed age. This analysis recomputed them in 719 people with entered pump settings and 1,678 people with device data, drawn from six Jaeb Center trials and the OpenAPS Data Commons, spanning ages 2 to 82 and daily doses of 7 to 107 U. Pooled across all ages the sensitivity constant was 2108 (95% CI 2048 to 2183), which excludes 1700. Restricted to adults it was 1915 (1828 to 1986) and in people under 18 it was 2390 (2281 to 2526). The pooled log-log slope of sensitivity on daily dose was -0.98 (-1.03 to -0.93), apparently confirming the inverse rule, but the same people restricted to adults gave -0.73 (-0.82 to -0.65). Because 51% of the pooled sample was under 18, the apparent confirmation was an artefact of age composition. The carbohydrate ratio constant measured from 23,596 announced meals that ended within 20 mg/dL of their starting glucose was 415 (405 to 425), and entered divided by measured was 1.02, indicating no detectable bias. Basal share was 0.51 overall and depended on the automation in use rather than on the person, at 0.51 with no algorithm, 0.50 under one commercial system, 0.40 under a bionic pancreas and 0.63 under a do-it-yourself loop. The sensitivity constant could not be measured in the same way. In the single open-loop cohort a correction of 1 to 3.5 U produced 15 to 20 mg/dL of glucose fall per acting unit against 45 entered, and in every closed-loop cohort the estimate was near zero or negative because comparison nights had already been corrected through basal. Entered and measured sensitivity are different quantities, since a pump setting encodes the whole expected fall including spontaneous reversion of 21 to 30 mg/dL over four hours. Fitting ISF = K / TDD^b to predict the overnight fall on held-out periods gave K = 880 (844 to 918) at Walsh's exponent of 1.0 across 1,613 people, which is 46% of the entered constant and is a marginal rather than a total quantity. The constant at that exponent spans 423 to 1433 between cohorts and the best-fitting exponent spans 0.50 to 1.10, so a single universal constant is not supported.

Introduction

Initial insulin pump settings are conventionally derived from total daily dose using constants that long predate continuous glucose monitoring. The sensitivity factor, the fall in glucose expected from one unit of rapid-acting insulin, is taken as 1700 divided by the daily dose in the formulation attributed to Walsh, or 1800 in the variant attributed to Davidson [1,2]. The carbohydrate ratio is taken as 500 divided by the daily dose, and basal insulin is taken as half of it.

These rules remain in wide use. They are the starting point in pump initiation protocols, they are embedded in the onboarding of automated insulin delivery systems, and the sensitivity factor in particular has acquired a second role: in several automated systems it is not only a starting value but a live parameter that the algorithm scales when it adjusts dosing.

Two features of the evidence behind them are worth stating. The cohorts used for validation have generally been small and of mixed age, and the quantity validated has generally been the setting people had entered rather than the response measured from their glucose. Both matter. If sensitivity factors differ systematically between children and adults, then a constant estimated by pooling them describes neither group, and the apparent fit of an inverse-proportional form can be produced by between-group differences in level rather than by any within-person relationship. Separately, an entered setting records what somebody was advised or has settled on, which need not be what a unit of insulin does.

Public release of several large trial datasets, together with the OpenAPS Data Commons, makes it possible to address both points in the same analysis. This paper recomputes the three constants in 719 people with entered settings and 1,678 people with device data, tests whether the inverse-proportional form survives stratification by age, and asks how far each constant can be measured from behaviour rather than read from a settings file.

An earlier analysis by the author, conducted in May 2026 on 138 open-source loop users, found all three rules inconsistent with the data and reported a log-log slope of -0.43 where the rule implies -1. That cohort is included here unchanged, and one purpose of the present work is to establish whether its conclusions were a property of that population.

Methods

Data sources

Device data came from six trials released by the Jaeb Center for Health Research through its public diabetes dataset portal, and from three extractions of the OpenAPS Data Commons.

Cohort	System	People	CGM readings	Median days
Loop	DIY closed loop	851	89,203,827	416
REPLACE-BG	Pump and CGM, no algorithm	196	12,034,638	238
IOBP2	Bionic pancreas	343	7,891,499	102
PEDAP	Control-IQ, ages 2–5	99	6,622,228	298
DCLP3	Control-IQ, adolescent and adult	112	5,673,867	183
DCLP5	Control-IQ, ages 6–13	100	5,535,124	218
v5 Trio	DIY closed loop	22	not applicable	not applicable
v6 AAPS	DIY closed loop	19	not applicable	not applicable
v7 oref0	DIY closed loop	97	not applicable	not applicable

The corpus holds 132,467,585 CGM readings, 68,995,020 basal records, 5,154,053 bolus records and 1,617,910 carbohydrate entries. The three OpenAPS Data Commons extractions carry settings and summary statistics rather than the raw streams, so they contribute to the entered-settings analysis alone.

REPLACE-BG occupies a distinct position in this corpus and carries much of the argument below. It ran in 2015, before automated delivery was available to its participants, so insulin was delivered by a pump following a schedule the person had programmed, with corrections given by hand. It is the only cohort here in which a night without a correction is a night on which nothing responded to glucose.

Dates are de-identified in every Jaeb release, by a per-participant random shift in some studies and by rebasing to a common epoch in others. Time of day survives both transformations, which is what this analysis requires. Loop timestamps carry a fixed per-participant offset from UTC, so daylight saving places time of day one hour out for part of each year; the effect on window start hours between 23:00 and 03:00 is tolerable and is noted as a limitation.

Entered settings

Sensitivity factors and carbohydrate ratios were taken as the per-person median of the values recorded at the time of dosing. For Loop and REPLACE-BG these came from the bolus calculator record, and for DCLP5 and PEDAP from the pump settings captured on case report forms, which store up to ten daily segments each with a start and end time; segment values were combined by time weighting. DCLP3 and IOBP2 ship no settings file, and the bionic pancreas has no sensitivity factor to record by design, so both are absent from the entered analysis.

Sensitivity factors in the Loop and REPLACE-BG calculator records are stored in mmol/L per unit for almost all participants. A per-person rule converted values whose median fell below 20 by a factor of 18.018, which reclassified 112,776 of 112,780 Loop rows. Conversion was verified by checking that the resulting distribution fell within 10 to 400 mg/dL per unit.

The settings loader was validated against an independent stream: of 80,039 Loop calculator rows carrying a carbohydrate entry, 100.00% matched a record in the separately loaded carbohydrate table at the same timestamp and amount.

Daily dose

Total daily dose was measured from each person's own pump record rather than taken from a case report form, summing delivered basal and bolus insulin over calendar days on which the pump recorded continuously, and requiring at least 30 such days. Measured daily dose correlated with the clinician-recorded value at Spearman 0.92 in DCLP5 and 0.89 in PEDAP.

One field required correction. A column labelled as a seven-day basal total proved on inspection to be a daily total, which was established by regressing it on the measured daily basal delivery.

Overnight windows

The sensitivity analyses used four-hour windows starting at 23:00, 00:00, 01:00, 02:00 and 03:00 local time. A window was retained when starting glucose lay between 90 and 300 mg/dL, no carbohydrate had been recorded for six hours before it, no bolus for three hours before it, glucose had not risen by more than 15 mg/dL in the preceding 30 minutes, and CGM coverage was complete. Screening conditions were evaluated only on information available before the window opened, so that no window was selected on the basis of its own outcome. In cohorts with no carbohydrate stream, fasting was inferred from the absence of any bolus large enough to be a meal dose, defined as 5% of that person's daily dose, which keeps the criterion free of scale.

Of 2,179,131 candidate windows, 708,106 passed screening, from 1,678 people.

Insulin action

Insulin action was computed by convolving delivery with the remaining-action curve of the appropriate model. For the Loop cohort the model is LoopKit's exponential form, with the peak and duration of the preset each participant was using, recovered by comparing recomputed insulin on board against the value the app recorded at each bolus; agreement reached $r = 0.927$. Elsewhere the oref exponential form with a six-hour duration and 75-minute peak was used.

Action within a window was separated into two components by convolution. The first is action arising from insulin delivered before the window opened, which is predetermined with respect to anything that happens inside it. The second is action from insulin delivered during the window, which is not.

Two unit errors in the underlying data required correction before any of this was reliable. Bolus durations were recorded in milliseconds rather than seconds for three studies, which placed insulin in 41,591 of one participant's 46,176 time bins; durations above 86,400 were rescaled and those implying more than eight hours were treated as instantaneous. Separately, timestamps returned at microsecond resolution were being cast as though they were nanoseconds, dividing every basal total by 1000 while leaving boluses correct. Both were caught by a mass conservation check comparing grid totals against database sums, which now agrees to 0.0000%.

Exposure and matching

The exposure for the sensitivity analysis is insulin delivered above or below the person's own scheduled basal profile, by whichever route it arrived. An algorithm running temporary basal above the schedule has given a correction, and one suspending delivery has given a negative correction. Defining the exposure as boluses alone would discard most of the dosing variation in every automated cohort. Where a study records a programmed schedule this was used directly; where it does not, the reference is that person's own rolling median delivery for each half hour of the day over the preceding 30 days.

Two estimators were applied. The first is a matched comparison: for each isolated overnight correction, comparison windows were drawn from the same person at the same hour with starting glucose within 15 mg/dL and no bolus, and sensitivity was taken as the difference in four-hour fall divided by the difference in acting insulin. Estimates were pooled as a ratio of sums across events rather than as a median of per-event ratios, because dividing each event by its own small and noisy denominator both inflates the tail and biases the centre toward zero.

The second is a stratified regression. Windows were grouped by person, hour, starting glucose to 15 mg/dL, slope into the window to 1 mg/dL per 5 min, and glucose one hour earlier to 25 mg/dL. Within each stratum the fall and the exposure were centred, and the pooled within-stratum slope was taken as the estimate. Strata with fewer than four windows or an exposure range below 0.4 U were dropped.

Measured carbohydrate ratio

For a meal that was announced and dosed for, and that left glucose where it started, the ratio that person required is the grams divided by the units, with no insulin model or basal assumption involved. Meals of at least 20 g with a dose within 20 minutes were selected, requiring no other carbohydrate for four hours before or five hours after and no further insulin during. Of 78,705 such meals from 1,011 people, 23,596 ended within 20 mg/dL of their starting glucose. Meals that did not end neutral were excluded rather than corrected, and people with fewer than eight qualifying meals were dropped, leaving 742.

Statistics

Intervals are percentile bootstrap over people, with 4,000 resamples for constants and 1,500 for slopes, resampling people rather than observations because windows from one person are not independent. Slopes are least squares on logarithms. Where per-person estimates were pooled, DerSimonian and Laird random effects pooling was used and heterogeneity is reported. Analysis ran locally in Python against a PostgreSQL 17.9 instance with TimescaleDB 2.26.1.

Results

Entered constants

Constant	Walsh	All 719	95% CI	Adults (422)	Under 18 (297)
Sensitivity x daily dose	1700	2108	2048–2183	1915	2390
Carb ratio x daily dose	500	409	392–422	404	414
Basal share of daily dose	0.50	0.49	0.48–0.50	0.52	0.47

The pooled sensitivity constant sits 24% above Walsh's value and its interval excludes it. Restricted to adults it falls to 1915 (1828 to 1986), which also excludes 1700. Within the adult group the two sources disagree substantially, at 1799 in the Jaeb trials against 2381 in the OpenAPS Data Commons, whose members set their own values over years rather than receiving them from a clinic. The one subgroup whose interval covers 1700 is Jaeb participants aged 45 and over, at 1766 (1680 to 1839).

The carbohydrate ratio constant misses 500 by approximately a fifth in every cohort and at every age. The basal rule holds, at 0.49 with an interval covering 0.50.

Age composition

Age band	n	Sensitivity x dose	95% CI	Covers 1700
2 to 5	115	2163	2045–2389	no
6 to 12	128	2561	2406–2695	no
13 to 17	54	2350	2243–2624	no
18 to 44	162	1830	1738–1944	no
45 and over	122	1766	1680–1839	yes

The constant falls from age six onward and reaches Walsh's value only in the oldest group. A child of ten carries a sensitivity constant approximately 45% above an adult of 45, which in dosing terms means the rule would correct that child considerably harder than their own settings prescribe.

The carbohydrate ratio constant moves in the opposite direction and more weakly, from 297 at ages 2 to 5, to 479 at 6 to 12, settling near 400 in adults. Only the school-age bands are consistent with 500.

The parametric form

Walsh's rule implies that sensitivity is inversely proportional to daily dose, which is a log-log slope of exactly -1.

Cohort	n	Slope	95% CI	Excludes -1
All Jaeb pooled	581	-0.98	-1.03 to -0.93	no
Jaeb adults only	284	-0.73	-0.82 to -0.65	yes
Jaeb under 18	297	-0.91	-0.96 to -0.85	yes
OpenAPS Data Commons	138	-0.43	-0.59 to -0.27	yes
Everything	719	-0.87	-0.93 to -0.82	yes

The pooled Jaeb slope of -0.98 appears to confirm the inverse rule. Restricting the same people to adults moves it to -0.73, and the interval then excludes -1. What produces the value near -1 is the mixing of young children, who use little insulin and carry high sensitivity factors, with adults who do neither. Between-group differences in level generate a gradient across the pooled sample that need not reflect how sensitivity varies within any individual.

This is not attributable to the range of daily doses sampled. The v7 and Loop cohorts span an identical spread of 1.43 in logs and return slopes of -0.46 and -0.94 respectively.

A validation of the 1700 rule performed on a cohort of mixed age will therefore tend to confirm it, and the confirmation will reflect composition. This applies to the author's May 2026 analysis as much as to any other.

Results by cohort

Cohort	n	Sensitivity x dose	Carb ratio x dose	Basal share	Slope
Loop	194	2043 (1913–2249)	375 (357–407)	0.58	-0.94
REPLACE-BG	192	1853 (1791–1936)	431 (410–460)	0.51	-0.72
DCLP5	100	2429 (2306–2580)	513 (479–542)	0.46	-1.00
PEDAP	95	2162 (2045–2317)	291 (266–317)	0.41	-0.89
v5 Trio	22	2682 (2216–2962)	373 (279–459)	0.45	-0.53
v6 AAPS	19		526 (409–607)	0.39	-0.03

Cohort	n	Sensitivity x dose	Carb ratio x dose	Basal share	Slope
		2953 (2201–5713)			
v7 oref0	97	2164 (2044–2566)	387 (315–436)	0.48	-0.46

The paediatric cohorts differ from one another more than they differ from the adult cohorts. PEDAP, covering ages 2 to 5, carries both the lowest carbohydrate ratio constant at 291 and the lowest basal share at 0.41. DCLP5, covering ages 6 to 13, carries the highest paediatric sensitivity constant at 2429 and a carbohydrate ratio constant of 513 whose interval covers 500.

The three OpenAPS Data Commons cohorts return shallow slopes, between -0.03 and -0.53, against -0.72 to -1.00 in the Jaeb cohorts. These are people who have adjusted their own settings over years, against people whose settings were set within a trial protocol. The rule fits best where it is followed.

Measured carbohydrate ratio

Cohort	n	Carb ratio x dose	95% CI	Covers 500
Loop	594	418	408–429	no
REPLACE-BG	148	390	362–427	no
Everyone	742	415	405–425	no

Among the 289 people who also had an entered carbohydrate ratio, entered divided by measured was 1.02. The estimator therefore carries no detectable bias, and 415 can be read as the constant the rule should use rather than as a description of what people have entered. It agrees closely with the entered figure of 409.

Basal share

Delivered basal over delivered total requires no model, so it covers the two cohorts with no settings file, adding 438 people omitted from the tables above.

Cohort	System	n	Basal share	Covers 0.50
Loop	DIY closed loop	830	0.60	no
REPLACE-BG	None	196	0.51	yes
DCLP3	Control-IQ	112	0.48	yes
DCLP5	Control-IQ	100	0.46	no

Cohort	System	n	Basal share	Covers 0.50
PEDAP	Control-IQ	95	0.41	no
IOBP2	Bionic pancreas	326	0.39	no
Everyone	Mixed	1,659	0.51	no

Restricting to adults removes age as an explanation and sharpens the pattern. REPLACE-BG, where no algorithm was running, sits at 0.51 and Control-IQ at 0.50, both consistent with Walsh. The bionic pancreas delivers 0.40 and the do-it-yourself loop 0.63. Walsh's half therefore describes what people programme when nothing is adjusting it, and survives one commercial system unchanged. The remaining two move it substantially and in opposite directions, which is a property of the algorithm rather than of the person using it.

Measured sensitivity

Cohort	Comparison	Events	Matched fall, 1–3.5 U	Per acting unit	Entered
REPLACE-BG	Open loop	239	22 to 35 mg/dL	15 to 20	45
Loop	DIY closed loop	269	-4 to 5	-2 to 5	57
DCLP3	Control-IQ	79	-8 to -4	-5 to -4	not recorded
DCLP5	Control-IQ	38	-7	-9	68
PEDAP	Control-IQ	50	-6	-11	161

Only the open-loop cohort returns a positive estimate. The explanation lies in the comparison nights rather than in the treated ones. The four-hour fall on nights when no correction was given, by starting glucose, was as follows.

Cohort	120–150	150–180	180–220	220–260	260+
REPLACE-BG	12	21	30	35	43
Loop	20	40	62	85	104
DCLP5	11	37	68	102	126
PEDAP	11	38	72	111	146

Under an algorithm a night without a bolus is not an untreated night. Glucose fell two to three times as far as in the open-loop cohort, because the algorithm delivered the correction through basal and arrived at a comparable endpoint. An added bolus therefore has little left to produce and little to measure.

This is not a consequence of the matching being too loose, and tightening it does not help. Progressive conditioning in the Loop cohort moved the estimate from 11.6 mg/dL per unit with person and hour alone, to 4.3 adding glucose to 50 mg/dL, to 2.8 at 15 mg/dL, and then to a plateau: 2.6 adding slope, 2.9 adding glucose one hour earlier, 2.6 tightening that further. A plateau argues against measurement error in the reconstructed exposure, which would continue to drive the slope toward zero as conditioning removed genuine variation. The large first step is consistent with removal of mean reversion, since glucose that is high both attracts insulin and falls without it.

What remains is that the algorithm's dose is tied to need the CGM trace does not carry. Within an identical stratum, the correlation between insulin committed before a window and insulin delivered during it was 0.054 in REPLACE-BG and 0.447 in Loop, with intermediate values of 0.062 in DCLP3, 0.140 in DCLP5, 0.151 in PEDAP and 0.212 in IOBP2. Someone who gives a correction and sleeps generates almost none of this association. A controller sampling every five minutes continues to respond to information the trace does not contain, so conditioning on the trace cannot break the association between its dosing and the outcome.

In the one cohort where the question can be posed, a unit produced 15 to 20 mg/dL of fall overnight against 45 entered, which as a constant is 630 to 840 against 1886. The ratio of measured to entered lay between 0.39 and 0.50 across four cohorts.

The constant fitted from observed response

The matched estimator above answers a narrow question, whether one additional unit can be shown to have done something, and under an algorithm it cannot. That is a property of that contrast rather than of the data. A constant can still be fitted, by asking which law of the form ISF = K / TDD^b best predicts the overnight fall on periods the fit never saw. Every person receives their own intercept, since overnight most insulin is basal offsetting hepatic glucose output and no sensitivity factor should be asked to carry that. Scoring is a per-person 70/30 split in time with median absolute error on the held-out tail.

Across 1,613 people the constant at Walsh's exponent of 1.0 was 880 (95% CI 844 to 918) with a held-out error of 23.41 mg/dL. The best-fitting exponent was 0.90 with a constant of 616 and an error of 23.39 mg/dL. Walsh's inverse-proportional form therefore costs 0.02 mg/dL against the best fit in this corpus, which is no meaningful penalty, and the form can be retained.

Cohort	People	Best exponent	Constant there	Constant at b = 1	Held-out MAE
Loop	796	0.80	384	774	19.93
IOBP2	323	0.55	143	869	30.67
REPLACE-BG	187	1.10	623	423	29.26
DCLP3	112	0.50	169	1177	21.61
DCLP5	100	0.95	1203	1433	20.39
PEDAP	95	1.00	1233	1233	23.38
Pooled	1,613	0.90	616	880	23.41

Two readings of this table are needed together. A constant is obtainable from observational data, and the pooled figure of 880 is estimated with a narrow interval on a large sample and validated out of sample. A single universal constant is not supported by these cohorts: the value at b = 1 spans 423 to 1433, a factor of 3.39, and the best exponent spans 0.50 to 1.10. The open-loop cohort, in which the matched estimator was the only one to work, returns the lowest value at 423, and its agreement with the matched estimate of 399 for the same people is the one place two independent methods can be checked against each other.

The pooled constant of 880 is 46% of the entered constant of 1915 for adults. That ratio is consistent with every other route taken in this paper, which found measured-to-entered ratios between 0.39 and 0.50.

The reason for the gap follows from the design rather than from error in either number. The fit gives each person an intercept, so 880 is the marginal fall per acting unit relative to that person's own baseline trajectory. A settings file encodes the whole fall a person expects after a correction, and part of that fall occurs without insulin: overnight and carbohydrate-free between 150 and 250 mg/dL, glucose fell a median 21 to 30 mg/dL over four hours with no correction given. Comparing 880 with 1700 or with 1915 without stating this is comparing a marginal quantity with a total one.

A scaling claim withdrawn

An earlier version of this analysis reported a different route to the same question, regressing a per-person measured constant on daily dose, and found a slope of +0.107 with an interval of -0.024 to +0.240. That was presented as evidence that one constant fits the whole dose range. It is withdrawn.

Two faults account for it. The regression was fitted only to the 1,182 people of 1,660 whose measured sensitivity came out above zero, which is selection on the outcome. More seriously, it pooled cohorts whose bias differs. Within-cohort slopes from that estimator run -0.053 in REPLACE-BG, -0.014 in IOBP2, +0.127 in DCLP3, +0.155 in PEDAP, +0.245 in Loop with an interval of +0.083 to +0.411 that excludes zero, and +0.323 in DCLP5. The pooled interval covered zero only by averaging across that spread.

The defence offered for the pooled figure was that a multiplier common to everyone cannot alter a slope. The multiplier is not common. The endogeneity measure above runs from 0.054 to 0.447 between cohorts, and cohort median daily dose spans 13.6 to 55.5 U, so a bias that differs by cohort differs along the axis the slope is measured on. This is the same composition artefact identified earlier in this paper for entered settings, arising in the author's own measured analysis. The predictive fit reported in the preceding section is the sounder route to the same question and reaches a compatible conclusion: the exponent is not stable across cohorts either, spanning 0.50 to 1.10.

Discussion

Four findings follow from this analysis.

The first concerns method, and it arose twice in this work. An inverse-proportional relationship between sensitivity and daily dose can be manufactured by pooling groups that differ in level, and in this corpus it was: the pooled slope of -0.98 became -0.73 in the same people restricted to adults, with 51% of the pooled sample under 18. Any validation of a dose-derived constant performed on a cohort of mixed age is therefore weak evidence for the functional form, and the direction of the artefact is toward confirmation.

The same artefact then appeared in the measured analysis, where the author initially reported a pooled slope near zero as evidence that one constant fits the whole dose range. It does not survive stratification. Within-cohort slopes from that estimator run from -0.05 to +0.32 and the largest cohort excludes zero at +0.245. The independent predictive fit reaches the same conclusion by a sounder route, since the best-fitting exponent there spans 0.50 to 1.10 between cohorts and the constant at a fixed exponent spans a factor of 3.39.

Pooling estimates whose bias differs by group, across groups that differ in the exposure, reproduces the error in a second place. This applies to the author's earlier work and to an earlier draft of the present paper.

The second concerns the constants themselves. The carbohydrate ratio rule is the clearest departure from practice and the most consistent, at 415 measured and 409 entered against 500, with agreement between the two approaches and no detectable bias in the measurement. The basal rule holds where nothing is adjusting delivery and does not describe what an automated system delivers, which is unsurprising once stated but is worth separating from any claim about the person. The sensitivity rule is 13% out in adults and 41% out in children, in the same direction, and covers 1700 only in adults aged 45 and over.

The third concerns what the sensitivity factor is, and it is the reason two defensible constants differ by a factor of two. A pump setting encodes the whole glucose fall a person expects after a correction. Part of that fall occurs without insulin: overnight and carbohydrate-free, between 150 and 250 mg/dL, glucose fell a median 21 to 30 mg/dL over four hours with no correction given. Any estimator that separates the insulin effect from that baseline returns a marginal quantity, and both estimators used here do. The predictive fit gives 880 divided by daily dose across 1,613 people and the matched design gives 15 to 20 mg/dL per unit against 45 entered in the open-loop cohort, which agree with each other and are each about 46% of the entered figure.

These are therefore two quantities rather than two estimates of one. Substituting the marginal value into a pump would roughly double correction doses. The entered constant of 1915 in adults remains the number to set, and 880 describes what an additional unit does against a person's own baseline. Reporting either without naming which one it is has caused confusion in this work and would cause it in anybody else's.

The fourth concerns study design. The absence of a measurable bolus effect in four automated cohorts is a finding about those systems rather than a failure of measurement, since the algorithm reached a comparable glucose endpoint through basal adjustment alone. Observational estimation of insulin sensitivity from automated insulin delivery data is obstructed by the controller acting on information that is not recoverable from the CGM trace, and the strength of that obstruction can be quantified by the association reported above. This bears on the growing literature that estimates sensitivity from closed-loop device data, and suggests such estimates require an open-loop comparison or an experimental manipulation of dose.

Limitations

The analysis is observational throughout, and no dose was assigned by the investigator. Confounding by indication is the dominant threat to the sensitivity estimates and is the explicit subject of the closed-loop results.

The window horizon is four hours against insulin models of six, so part of the glucose response falls outside the observation. Several mechanisms could each produce a measured-to-entered ratio near 0.4. Truncation at four hours is one. Selection into correcting at times when glucose is expected to remain high is a second, and optimism in entered settings a third. The present design does not separate them, and extending the horizon to six hours would address the first. That extension is planned.

The measured sensitivity estimator returns a value at or below zero for 29% of people, 478 of 1,660, and any analysis restricted to positive values is selecting on the outcome. Survival of that filter is close to independent of daily dose, at Spearman +0.042 with $p = 0.084$, so it does not by itself generate a spurious scaling. It does mean the surviving group is not a random sample of the cohort, and the pooled estimates in the measured section should be read with that in mind.

Insulin action is reconstructed rather than observed. For the Loop cohort the model was selected per person against the app's own recorded insulin on board, reaching $r = 0.927$, but elsewhere a single model was assumed.

Entered settings were available for 719 people of the 1,678 with device data, and two cohorts ship no settings file. The under-18 entered figures rest on Jaeb participants alone, since the OpenAPS Data Commons cohort contains no children.

Daylight saving displaces time of day by one hour for part of the year in the Loop cohort. Body weight is not available in the loaded data, so sensitivity per kilogram could not be examined, and weight is a plausible mediator of the age pattern reported here. The OpenAPS Data Commons participants are self-selected and technically engaged, which is consistent with their distinct slopes and should temper generalisation from them.

Conclusions

Recomputed across 719 people with entered settings and 1,678 with device data, the carbohydrate ratio constant is 415 rather than 500 and the basal share is 0.51 where no algorithm is adjusting delivery. The sensitivity constant is 1915 in adults and 2390 in people under 18, and pooling those groups produces both a constant and a functional form that describe neither. The inverse-proportional form is not supported once age is held constant. A sensitivity constant can be fitted from observed glucose response, at 880 divided by daily dose across 1,613 people, and it predicts held-out overnight falls to 23.4 mg/dL. It is 46% of the constant people enter, because it is the marginal fall per unit against each person's own baseline while a settings file encodes the total expected fall. It is not universal, spanning 423 to 1433 between cohorts at a fixed exponent, so it is a population starting value rather than a law.

Data availability

The Jaeb datasets are available from https://public.jaeb.org/datasets/diabetes under the terms stated there. The OpenAPS Data Commons is available on application to its curators. No participant-level data are redistributed with this work.

Code availability

Analysis code is at https://github.com/tim2000s/dynamic-isf-calculations. The entered constants are produced by `inv009/walsh_constants.py`, the measured carbohydrate ratio and basal share by `inv009/measured_basal_cr.py`, the per-cohort sensitivity measurement by `inv009/measured_isf_by_study.py`, and the net-dose estimator and its diagnostics by `inv009/net_dose_isf.py`.

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